NETWORKING

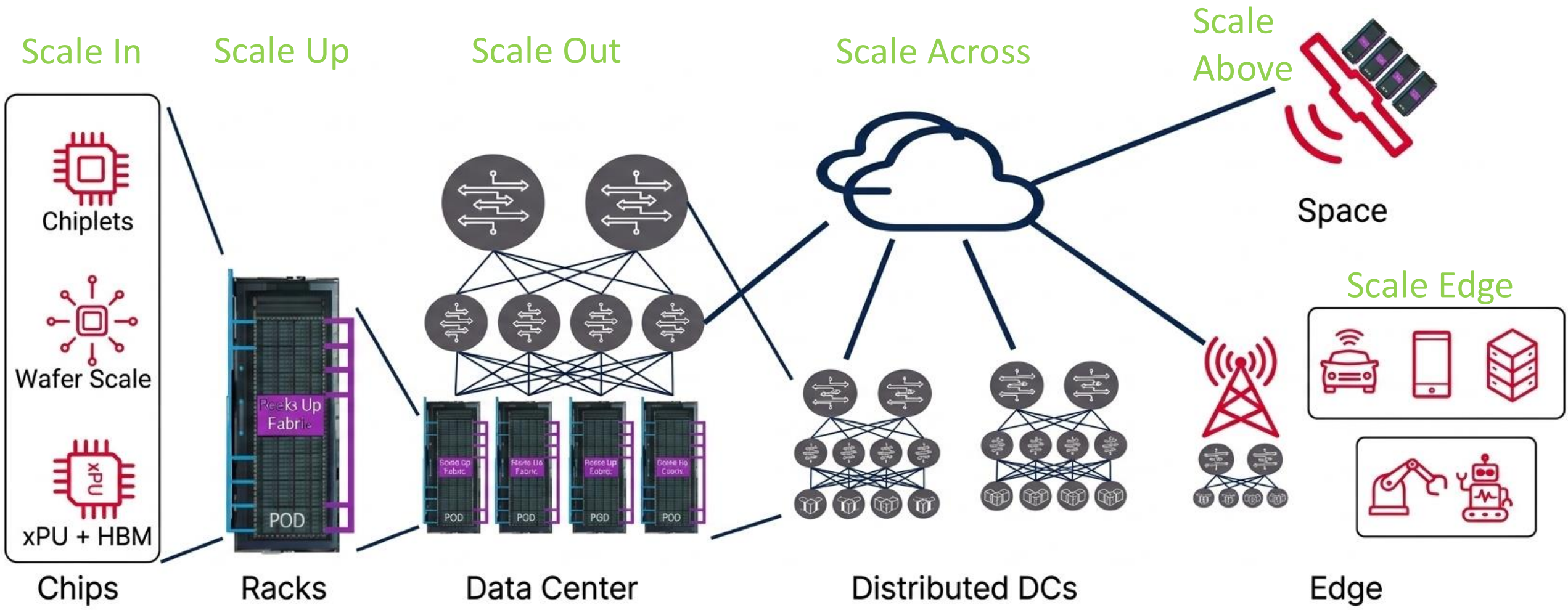
OCP ESUN Validation Methodology

Venkat Pullela *CTO, AI Infrastructure, Keysight*

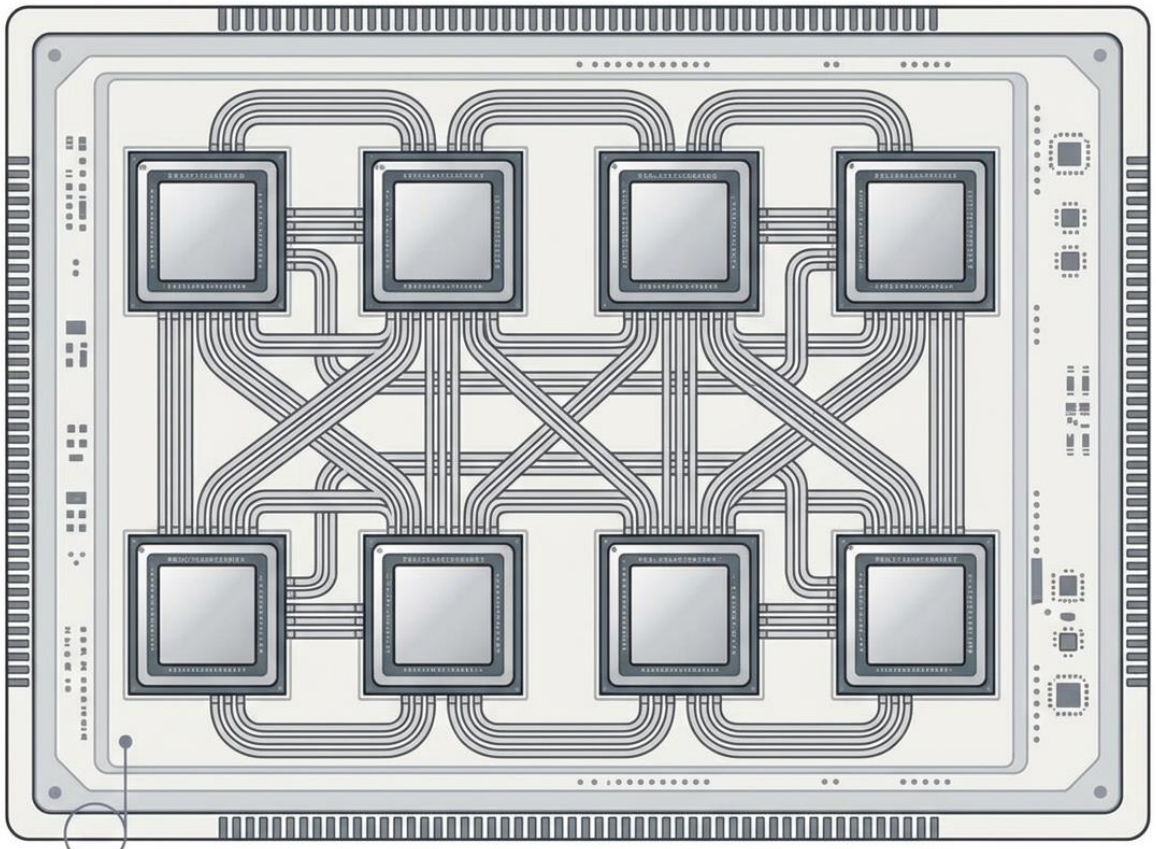
Manoj Wadekar *AI Systems Technologist, Meta*

OCP PLATINUM Member

Scaling AI Infrastructure



AI Scale Up Networks



Multiple GPUs fused into a single Unified Virtual Accelerator across a shared memory domain

- Tightly coupled – within a rack/pod
- Single hop or highly flattened topology
- Direct GPU-to-GPU memory sharing & collective operations

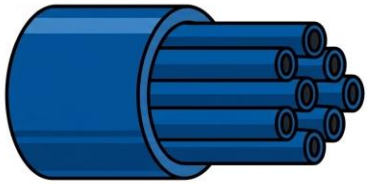
Sub-Microsecond Latency



Lossless Transport

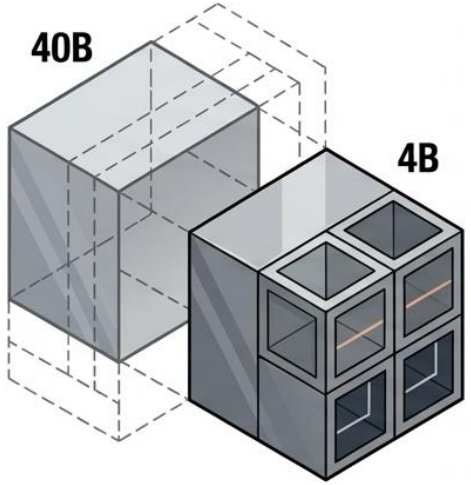


Massive Bandwidth Density



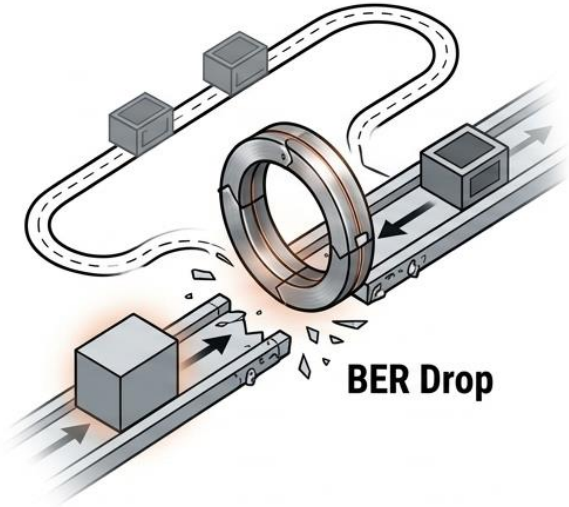
ESUN Architecture

Transforming standard Ethernet into a lossless, sub-microsecond fabric optimized for synchronized GPU clusters



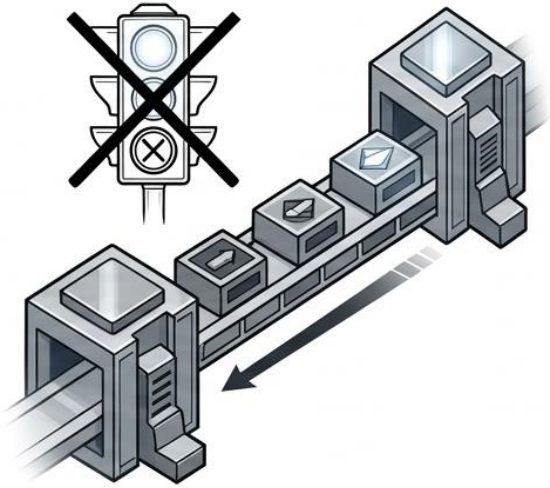
4-Byte Header

Compact header to maximize goodput



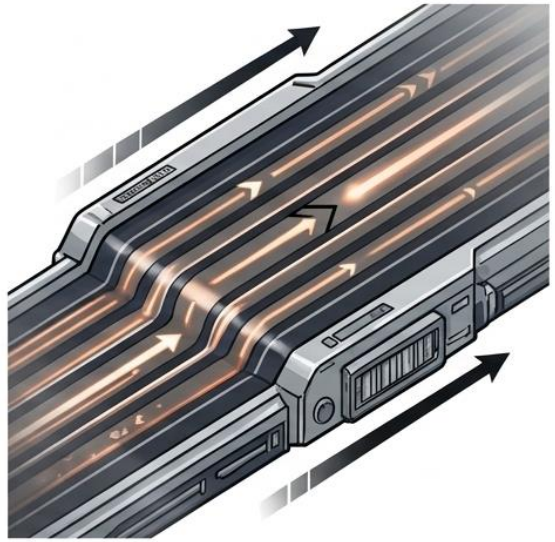
Link Level Retry (LLR)

UEC defined feature to reduce tail latency



Credit Based Flow Control (CBFC)

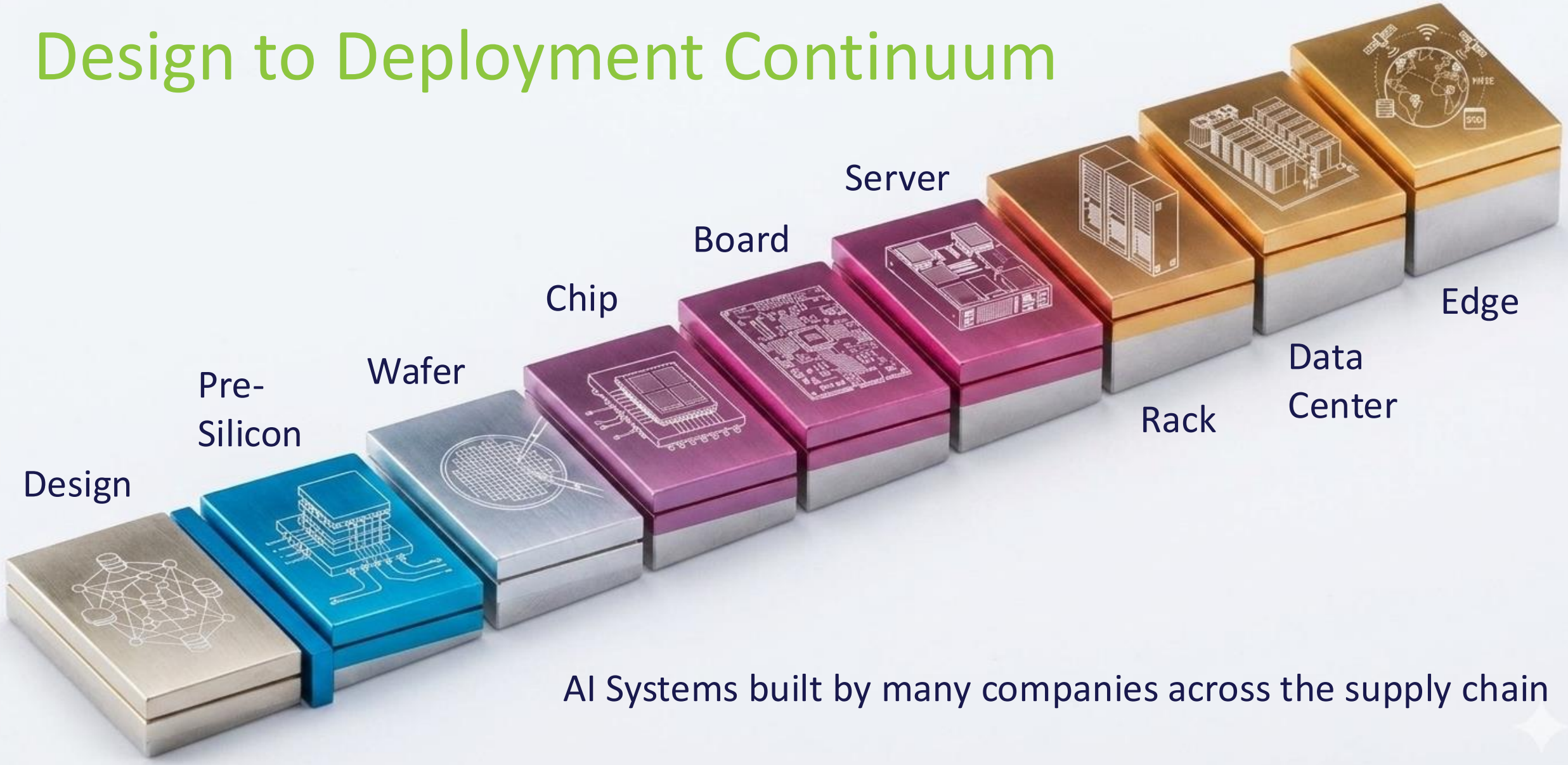
Lossless network without PFC



224G PHY

Sub-microsecond deterministic latency

Design to Deployment Continuum



AI Systems built by many companies across the supply chain

AI System Development Process

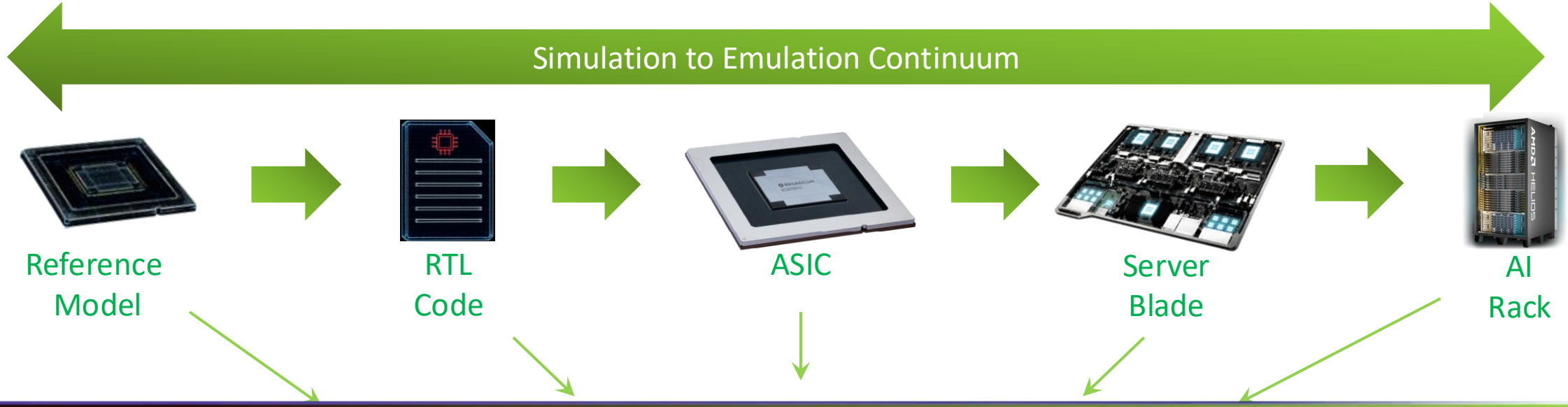
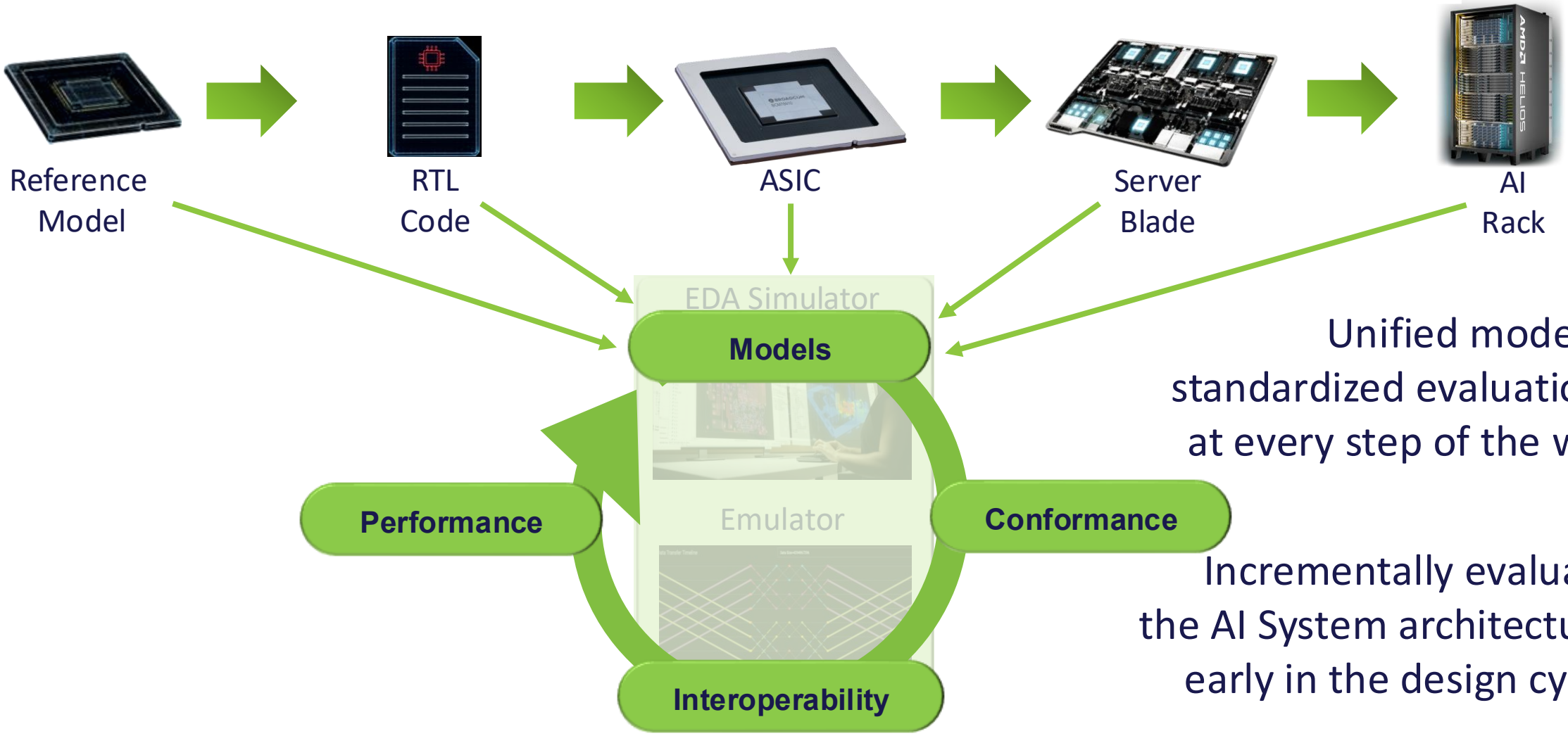


Image credits: NVIDIA, AMD, Broadcom

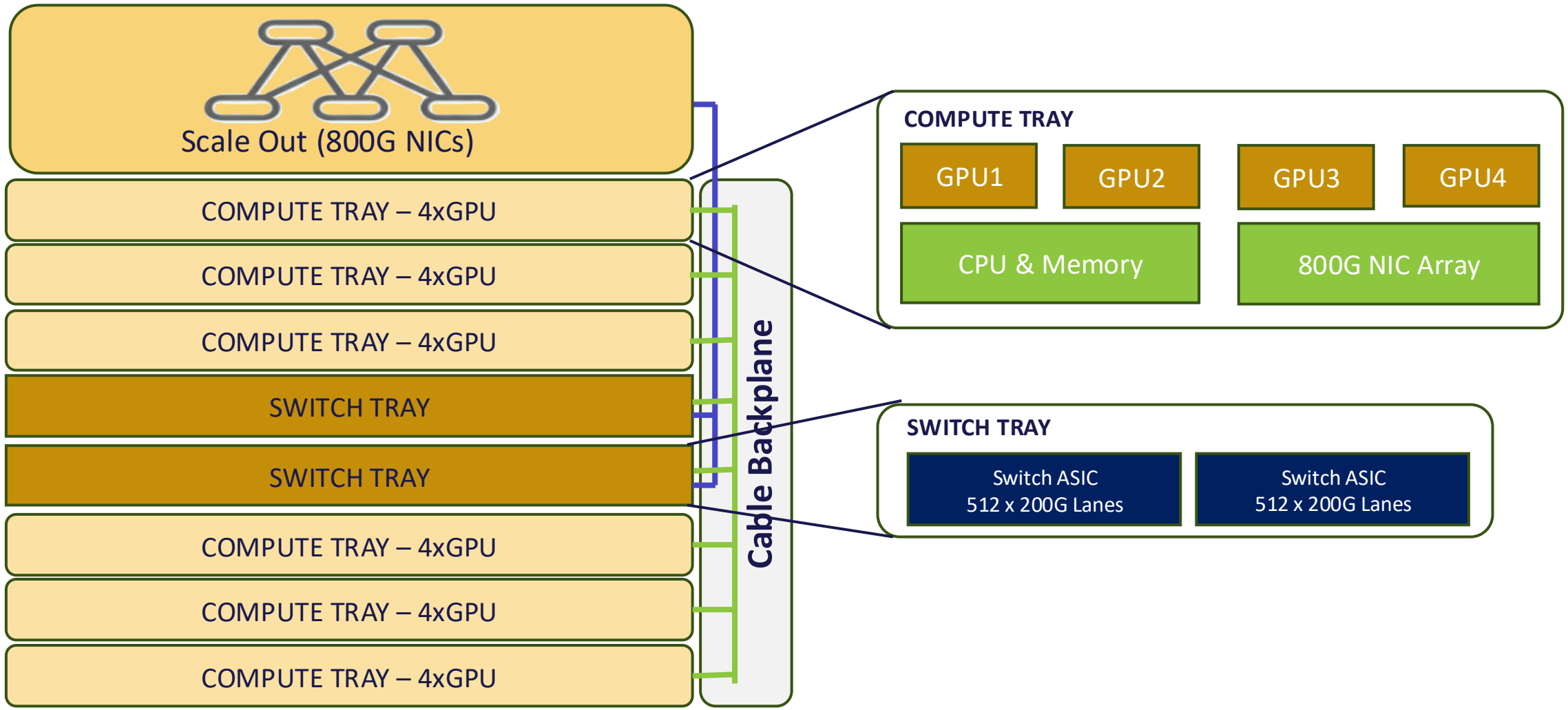
AI System Design and Development



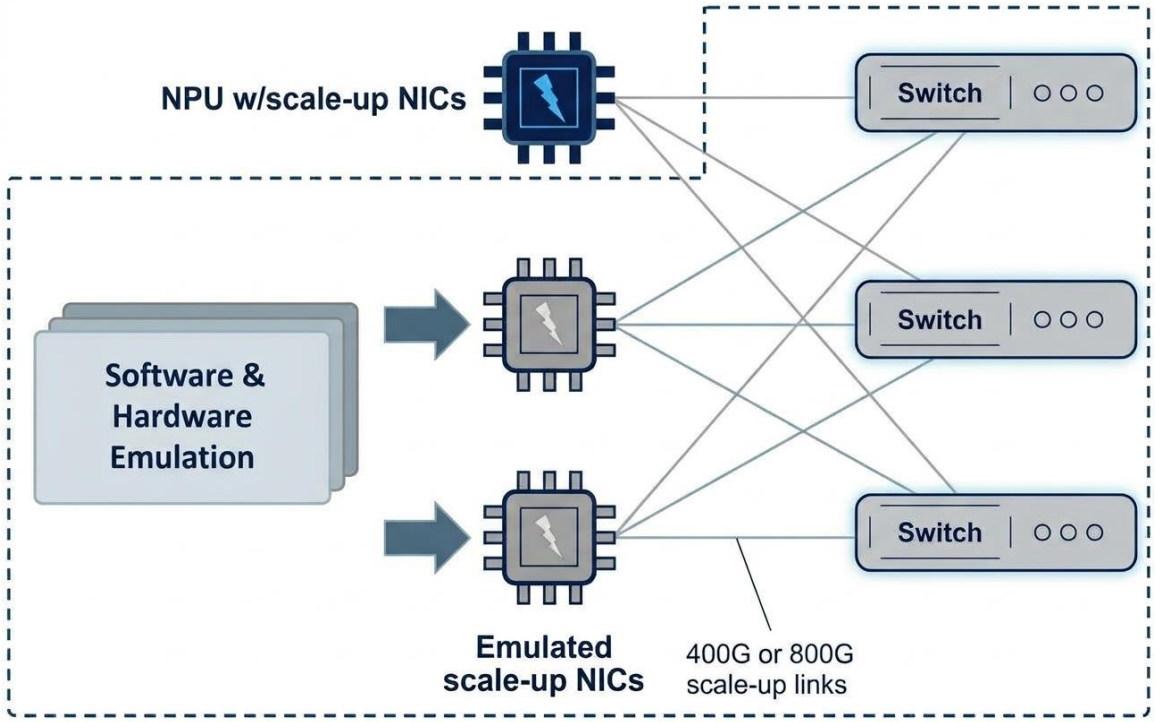
Unified models, standardized evaluations at every step of the way

Incrementally evaluate the AI System architecture early in the design cycle

AI Rack

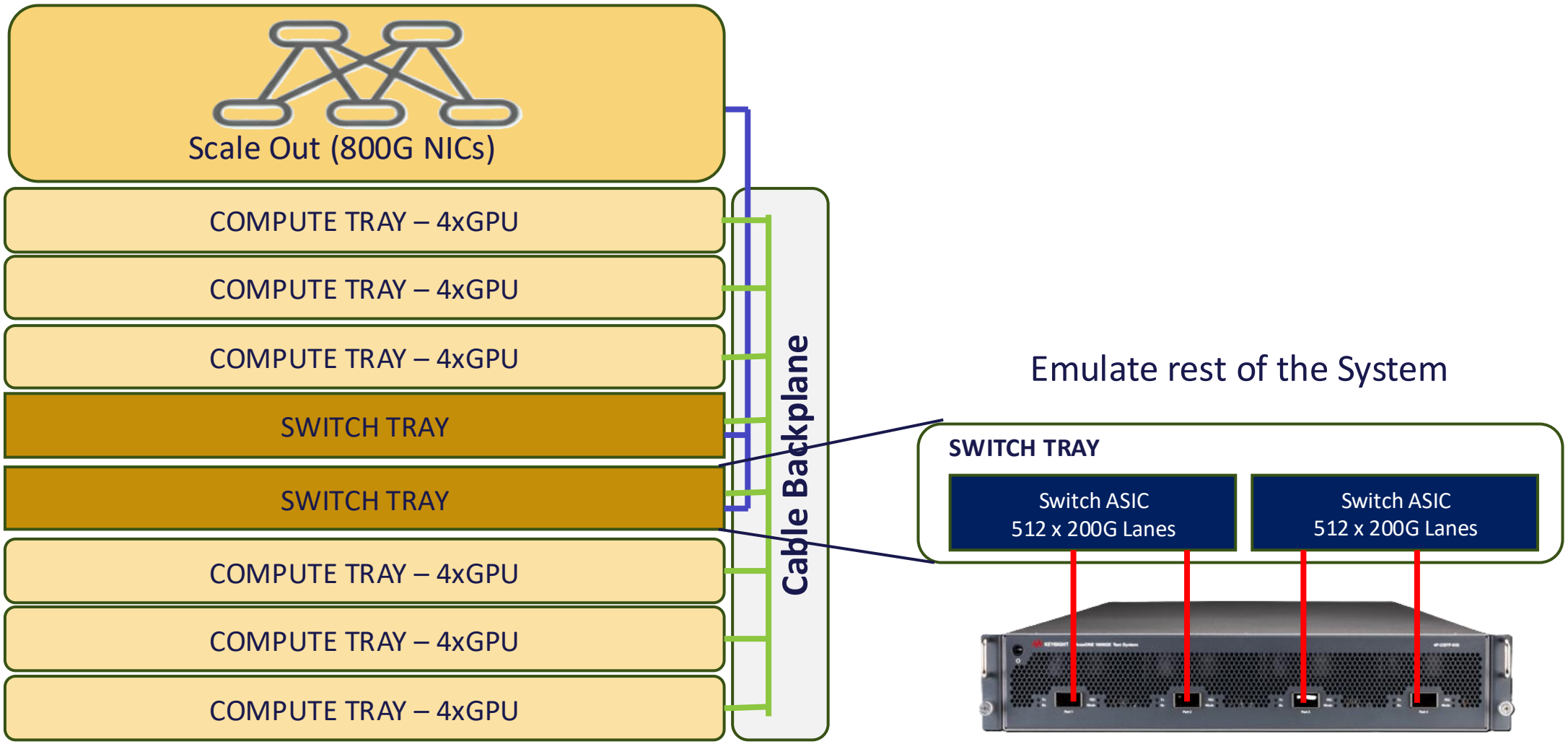

Scale Out (800G NICs)
COMPUTE TRAY – 4xGPU
COMPUTE TRAY – 4xGPU
COMPUTE TRAY – 4xGPU
SWITCH TRAY
SWITCH TRAY
COMPUTE TRAY – 4xGPU
COMPUTE TRAY – 4xGPU
COMPUTE TRAY – 4xGPU
Cable Backplane
COMPUTE TRAY
GPU1
GPU2
GPU3
GPU4
CPU & Memory
800G NIC Array
SWITCH TRAY
Switch ASIC
512 x 200G Lanes
Switch ASIC
512 x 200G Lanes

Accelerator/Switch ASIC Validation

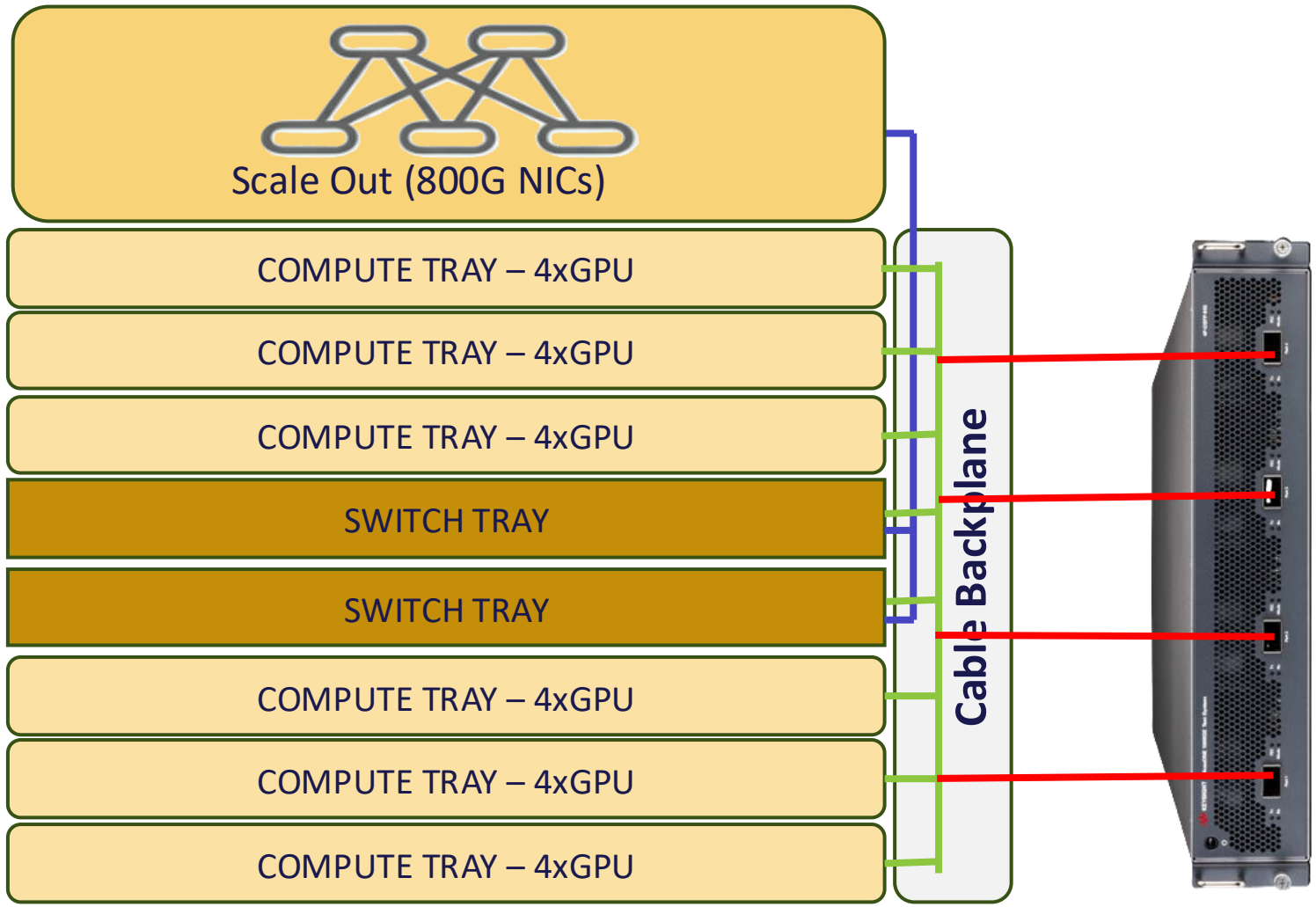


- Surround ASICs with emulated components
- Validate functionality
- Conformance to standards
- Interoperability
- Incrementally Emulate rest of the Rack

Switch tray Validation

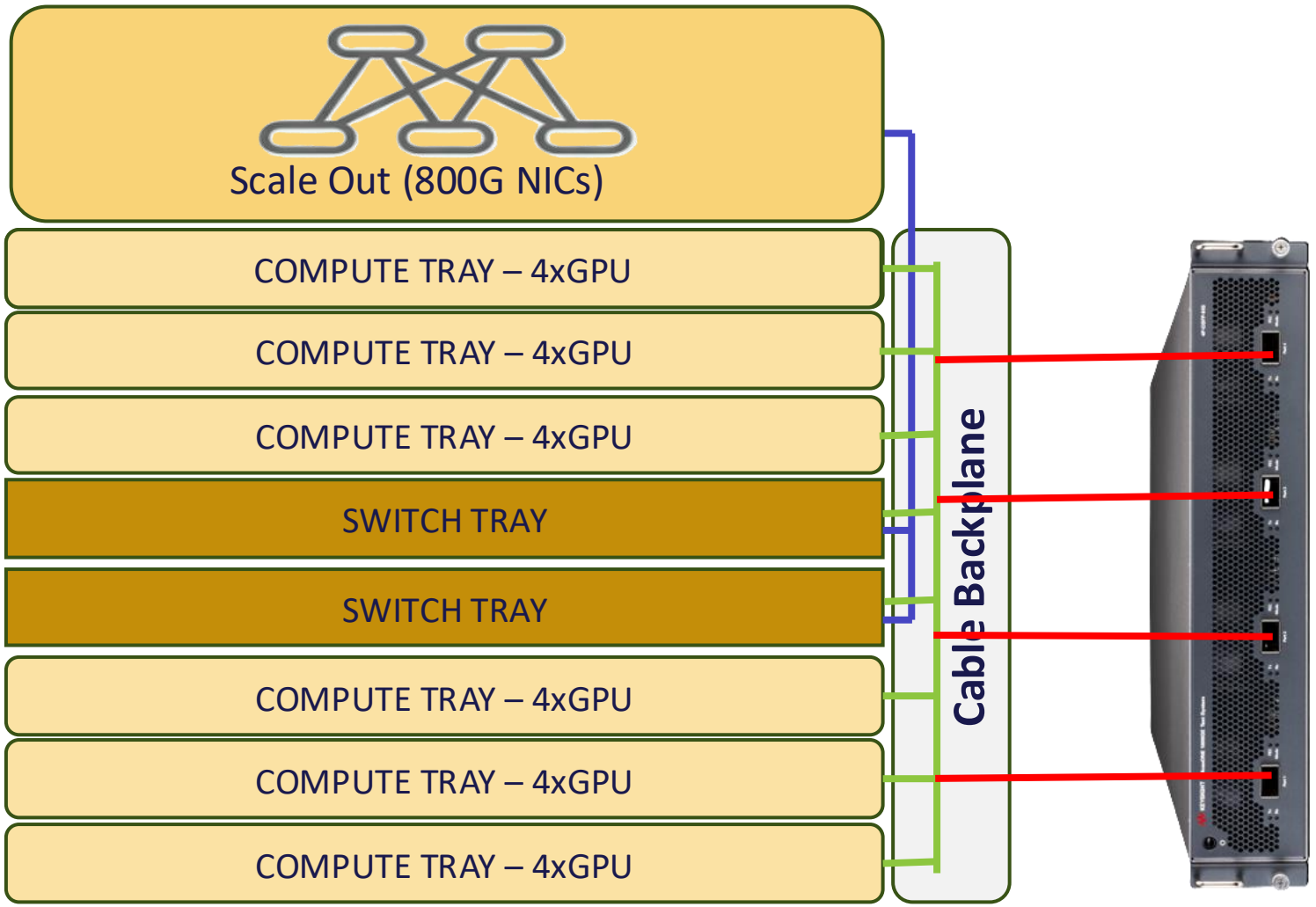


Backplane Validation



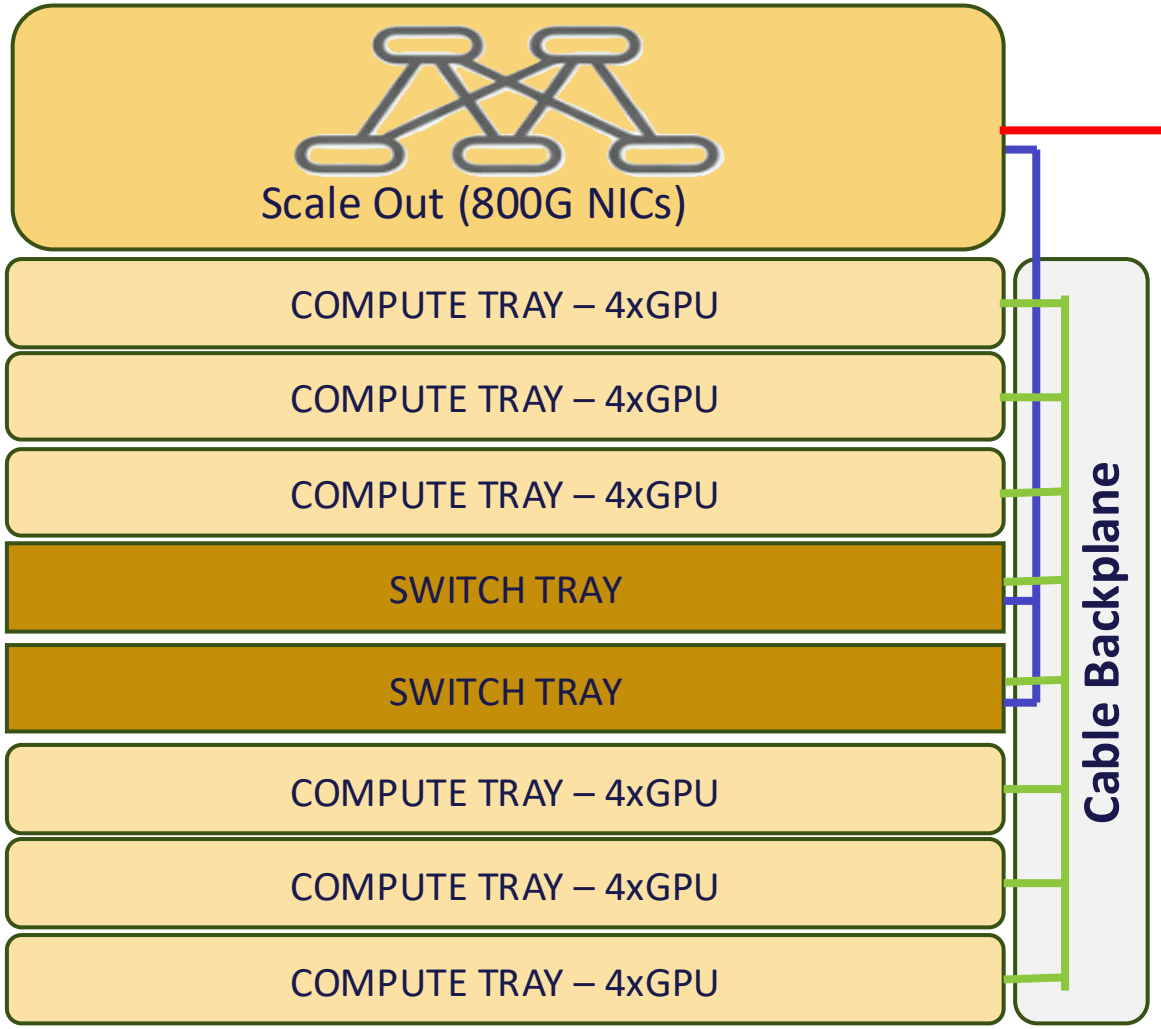
Emulate Compute trays

Compute tray Validation



Emulate rest of the compute trays

Scale Out Validation





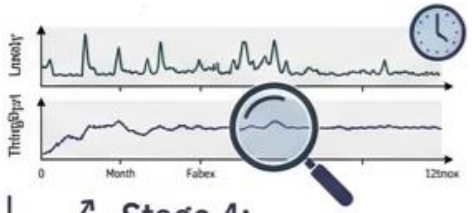
Emulate other racks

Workload Validation









	Stage 1: Baseline	Stage 2: Workload Realism	Stage 3: Shared Contention	Stage 4: Stability Over Time
Core Question	Perform as expected in isolation?	Does realistic structure alter isolated behavior?	Does shared contention alter workload behavior?	Is the shared environment stable long-term?
Minimum Condition	Single collective, isolated workload, short interval.	Isolated workload; representative phases, collective mix, and timing.	Concurrent workloads sharing a single contention domain.	Stage 3 conditions extended for a sustained duration.
Establishes	Baseline path health and isolated collective performance.	Impact of workload structure on isolated performance.	Cross-workload interference effects on target workload.	Variance growth, tail latency widening, recurring congestion.
Still Unknown	Behavior under representative workload structure.	Behavior under shared multi-tenant contention.	Long-term persistence or transience of interference.	Exact hardware-, queue-, or policy-level root causes.

Increasing Observability Granularity

Coarse per-collective metrics	Per-collective, per-phase visibility	Tenant-aware, flow-aware attribution	Per-iteration, time-aware analysis

Call to action

- Develop community standard validation methodology
- Create a separate working stream for Debug/Conformance/Performance
- Where to find additional information
  - ESUN Specification : https://www.opencompute.org/documents/ocp-esun-network-operator-requirements-base-specification-rev-1-0-final-pdf
  - ESUN Wiki: https://www.opencompute.org/wiki/Networking/ESUN
  - SONiC Scale UP Working group: https://lists.sonicfoundation.dev/g/SONiC-Scale-Up-WG/subgroups

Thank You!

